LECTURE 01 - GENERALITIES ON LIE ALGEBRAS, SOLVABLE AND NILPOTENT LIE ALGEBRAS

FANGYANG TIAN

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This series of lecture notes is devoted to the course *Lie Groups and Lie Algebras* I taught in Zhejiang University in Spring 2024. This series has a large overlap with the one I used in Spring 2023, with some modifications. Based on past teaching experience, I guess a better way is to pace through basic Lie theory steadily without lingering much on technical details. For instance, some foundational results, such as basic properties of Lie subgroups, are proved purely via a differential geometric approach, hence proofs are unrelated to the main subject of this course and hence will be completely omitted; while others, such as various versions of Schur's lemma, need prerequisites in functional analysis or universal enveloping algebra. These technical proofs will be postponed to later lectures. Students are encouraged to only keep these results in mind at first and review their proofs when are fully prepared later. All the materials presented in this series are based on several standard textbooks on Lie theory, with proofs occasionally given by myself. If any student spots anything suspicious (typo or even math mistakes), please feel free to contact me.

Unless further notice, **we always assume that the ground field F is of characteristic zero**. Even main theorems may fail in the positive characteristic case.

1. Lie Algebras, Homomorphisms and Field Extensions

Definition 1.1. *A **Lie algebra** over a field F is an F-vector space equipped with a product (called **Lie bracket**) $[X, Y]$ such that*

(1) *the product is bilinear and skew-symmetric (i.e. $[X, Y] = -[Y, X]$);*
(2) *the **Jacobian identity** holds:*

$$[[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y] = 0.$$

Definition 1.2. *A subset $\mathfrak{h}$ of a Lie algebra $\mathfrak{g}$ is called a **Lie subalgebra** if it is a Lie algebra under the Lie bracket operation of $\mathfrak{g}$. We say $\mathfrak{h}$ is an **ideal** of $\mathfrak{g}$ if $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$. When*

$\mathfrak{h}$ is an ideal, the quotient $\mathfrak{g}/\mathfrak{h}$ becomes a Lie algebra naturally once equipped with the Lie bracket

$$[\overline{X}, \overline{Y}] := \overline{[X, Y]}.$$

Here $\mathfrak{g} \to \mathfrak{g}/\mathfrak{h}, \quad X \mapsto \overline{X}$ is the quotient map as usual.

Example 1.3. *The vector space of matrices $\mathfrak{gl}_n(F) := \mathrm{Mat}_n(F)$ over a field F equipped with the Lie bracket $[X, Y] := XY - YX$ is a Lie algebra over F. The subspace*

$$\mathfrak{sl}_n(F) := \{X \in \mathfrak{gl}_n(F) \,|\, \mathrm{Tr}(X) = 0\}$$

is a Lie subalgebra of $\mathfrak{gl}_n(F)$. Indeed, for any $X, Y \in \mathfrak{gl}_n(F)$,

$$\mathrm{Tr}([X, Y]) = \mathrm{Tr}(XY - YX) = 0.$$

Example 1.4. *Set J_n to be the $n \times n$ antidiagonal matrix defined inductively by*

$$J_n := \begin{pmatrix} & 1 \\ J_{n-1} & \end{pmatrix}.$$

Then

$$\mathfrak{sp}_{2n}(F) := \left\{ X \in \mathfrak{gl}_{2n}(F) \,\middle|\, X \cdot \begin{pmatrix} & J_n \\ -J_n & \end{pmatrix} + \begin{pmatrix} & J_n \\ -J_n & \end{pmatrix} \cdot X^t = 0 \right\}$$

is a Lie subalgebra of $\mathfrak{gl}_{2n}(F)$. A direct computation shows that $\mathfrak{sp}_{2n}(F)$ consists of matrices of the form

$$\begin{pmatrix} a & b \\ c & -J_n a^t J_n \end{pmatrix},$$

where $b^t = J_n b J_n, c^t = J_n c J_n$.

Indeed, if $X, Y \in \mathfrak{sp}_{2n}(F)$, we need to check that $XY - YX \in \mathfrak{sp}_{2n}(F)$. For simplicity, we set $J = \begin{pmatrix} & J_n \\ -J_n & \end{pmatrix}$. Then indeed

$$(XY - YX)J + J(XY - YX)^t = XYJ - YXJ + JY^t X^t - JX^t Y^t = 0.$$

Example 1.5. *Set J_n to be the $n \times n$ antidiagonal matrix defined inductively by*

$$J_n := \begin{pmatrix} & 1 \\ J_{n-1} & \end{pmatrix}.$$

Then

$$\mathfrak{so}_{2n}(F) := \left\{ X \in \mathfrak{gl}_{2n}(F) \,\middle|\, X \cdot \begin{pmatrix} & J_n \\ J_n & \end{pmatrix} + \begin{pmatrix} & J_n \\ J_n & \end{pmatrix} \cdot X^t = 0 \right\}$$

is a Lie subalgebra of $\mathfrak{gl}_{2n}(F)$. A direct computation shows that $\mathfrak{so}_{2n}(F)$ consists of matrices of the form

$$\begin{pmatrix} a & b \\ c & -J_n a^t J_n \end{pmatrix},$$

where $b^t = -J_n b J_n, c^t = -J_n c J_n$. When $F = \mathbb{R}$, we usually write it as $\mathfrak{so}(n, n)$.

The proof that $\mathfrak{so}_{2n}(F)$ is a Lie subalgebra is the same as Example 1.4.

Example 1.6. *Set J_n to be the $n \times n$ antidiagonal matrix defined inductively by*

$$J_n := \begin{pmatrix} & 1 \\ J_{n-1} & \end{pmatrix}.$$

Then

$$\mathfrak{so}_{2n+1}(F) := \left\{ X \in \mathfrak{gl}_{2n+1}(F) \,\middle|\, X \cdot \begin{pmatrix} & & J_n \\ & 1 & \\ J_n & & \end{pmatrix} + \begin{pmatrix} & & J_n \\ & 1 & \\ J_n & & \end{pmatrix} \cdot X^t = 0 \right\}$$

is a Lie subalgebra of $\mathfrak{gl}_{2n+1}(F)$. A direct computation shows that $\mathfrak{so}_{2n+1}(F)$ consists of matrices of the form

$$\begin{pmatrix} a & w & b \\ u^t & 0 & -w^t J_n \\ c & -J_n u & -J_n a^t J_n \end{pmatrix},$$

where $b^t = -J_n b J_n, c^t = -J_n c J_n$. When $F = \mathbb{R}$, we usually write it as $\mathfrak{so}(n+1, n)$.

The proof that $\mathfrak{so}_{2n+1}(F)$ is a Lie subalgebra is the same as Example 1.4.

Example 1.7. *The compact unitary Lie algebra*

$$\mathfrak{u}(n) := \{X \in \mathfrak{gl}_n(\mathbb{C}) \,|\, X + \overline{X}^t = 0\},$$

and

$$\mathfrak{su}(n) := \mathfrak{sl}_n(\mathbb{C}) \cap \mathfrak{u}(n).$$

The proof that $\mathfrak{u}(n)$ is a Lie subalgebra is the same as Example 1.4. The compact orthogonal Lie algebra

$$\mathfrak{o}(n) := \mathfrak{u}(n) \cap \mathfrak{gl}_n(\mathbb{R})$$

and

$$\mathfrak{so}(n) := \mathfrak{o}(n) \cap \mathfrak{sl}_n(\mathbb{R}).$$

Though in prior, the definition of $\mathfrak{so}(n)$ requires the trace of any matrix in $\mathfrak{so}(n)$ is zero, yet in fact, every matrix in $\mathfrak{o}(n)$ has trace zero. Why? This is because by definition

$$\mathfrak{o}(n) = \{X \in \mathfrak{gl}_n(\mathbb{R}) \,|\, X + X^t = 0\}.$$

By taking the trace, $2\mathrm{Tr}(X) = 0$ for all $X \in \mathfrak{o}(n)$. Hence $\mathfrak{o}(n) = \mathfrak{so}(n)$.

Remark 1.8. *The definition of symplectic Lie algebra and orthogonal Lie algebra given above are targeted to the application in representation theory (or related modern automorphic forms); for geometers, it is (perhaps) more common to use a conjugation of the above realization of symplectic and orthogonal Lie algebras. We will (hopefully) come back to this issue later in the course.*

Proposition 1.9 ([Kna02, Example 1.1.13, 1.1.14, Proposition 1.7, 1.21])**.**

(1) *Finite intersection, finite sum and Lie bracket of ideals of a Lie algebra $\mathfrak{g}$ are still ideals of $\mathfrak{g}$.*

(2) *For any subset $\mathfrak{c} \subseteq \mathfrak{g}$, the* ***centralizer***

$$\mathfrak{z}_{\mathfrak{g}}(\mathfrak{c}) := \{Z \in \mathfrak{g} \,|\, [Z, X] = 0, \text{ for all } X \in \mathfrak{c}\}$$

is a subalgebra of $\mathfrak{g}$. Moreover, when $\mathfrak{c} = \mathfrak{g}$, the center

$$\mathfrak{z}_{\mathfrak{g}} := \{Z \in \mathfrak{g} \,|\, [Z, X] = 0, \text{ for all } X \in \mathfrak{g}\}$$

is an ideal of $\mathfrak{g}$.

(3) *For any subalgebra* $\mathfrak{h} \subseteq \mathfrak{g}$*, the* ***normalizer***

$$\mathfrak{n}_{\mathfrak{g}}(\mathfrak{h}) := \{Z \in \mathfrak{g} \,|\, [Z, X] \in \mathfrak{h} \text{ for all } X \in \mathfrak{h}\}$$

is a Lie subalgebra of $\mathfrak{g}$*.*

(4) *The* ***commutator*** *or* ***derived subalgebra*** $[\mathfrak{g}, \mathfrak{g}]$ *is an ideal of* $\mathfrak{g}$*. (This follows directly from definition, check by yourselves.)*

(5) *The kernel of a Lie algebra homomorphism is an ideal of* $\mathfrak{g}$*.*

(6) *For any Lie algebra* $\mathfrak{g}$ *over* F*, the space of* ***derivations***

$$\mathrm{Der}_F(\mathfrak{g}) := \{D \in \mathrm{End}_F(\mathfrak{g}) \,|\, D \text{ satisfies the Leibniz rule } D[X, Y] = [DX, Y] + [X, DY]\}$$

forms a Lie algebra over F*.*

Definition 1.10. *Suppose* $\mathfrak{g}, \mathfrak{h}$ *are Lie algebras over* F*. A* ***Lie algebra homomorphism*** $\varphi : \mathfrak{g} \to \mathfrak{h}$ *is a linear map which preserves the Lie bracket operation. A* ***Lie algebra isomorphism*** *is a bijective Lie algebra homomorphism.*

Example 1.11. *Let* $\mathfrak{g}$ *be a Lie subalgebra of* $\mathfrak{gl}_n(\mathbb{C})$*, closed under negative conjugate transpose (such Lie algebra is called to be reductive, which will be discussed later). Define* $\theta : \mathfrak{g} \to \mathfrak{g}, X \mapsto -\overline{X}^t$*. Then* θ *is a Lie algebra automorphism. Indeed, it is obvious that* θ *is bijective. We check that* $\theta([X, Y]) = [\theta(X), \theta(Y)]$*:*

$$\theta([X, Y]) = \theta(XY - YX) = [-\overline{X}^t, -\overline{Y}^t] = [\theta(X), \theta(Y)].$$

Example 1.12. *Given a vector space* V *over* F*, the associative algebra* $End_F(V)$ *is a Lie algebra by setting* $[A, B] = AB - BA$*. A* ***representation*** *of* $\mathfrak{g}$ *is just a Lie algebra homomorphism* $\mathfrak{g} \to End_F(V)$*. In particular, the map* $\mathfrak{g} \to \mathrm{End}_F(\mathfrak{g})$ *which sends* X *to* $[X, \cdot]$ *is a representation, called the* ***adjoint representation****. A representation of* $\mathfrak{g}$ *is also called a* $\mathfrak{g}$*-module.*

Theorem 1.13. (1) *A surjective Lie algebra homomorphism* $\varphi : \mathfrak{g} \to \mathfrak{h}$ *descends an isomorphism of Lie algebras :*

$$\mathfrak{g}/\ker\varphi \simeq \mathfrak{h}.$$

(2) *Let* $\mathfrak{a}, \mathfrak{b}$ *be ideals of* $\mathfrak{g}$*. Then we have an isomorphism of Lie algebras:*

$$(\mathfrak{a} + \mathfrak{b})/\mathfrak{a} \simeq \mathfrak{b}/(\mathfrak{a} \cap \mathfrak{b})$$

Proof. (1) Let $\overline{\varphi}$ be the vector space isomorphism $\mathfrak{g}/\ker\varphi \to \mathfrak{h}$ induced by φ. We only need to check that $\overline{\varphi}$ preserves Lie bracket. As usual, we write $\mathfrak{g} \to \mathfrak{g}/\ker\varphi, \quad X \mapsto \overline{X}$ for the quotient map. Then it is trivial to check

$$\overline{\varphi}([\overline{X}, \overline{Y}]) = \overline{\varphi}(\overline{[X, Y]}) = \overline{\varphi([X, Y])} = \overline{[\varphi(X), \varphi(Y)]} = [\overline{\varphi}(\overline{X}), \overline{\varphi}(\overline{X})].$$

(2) We know that $\mathfrak{h} := \mathfrak{a} + \mathfrak{b}$ is an ideal of $\mathfrak{g}$ and $\mathfrak{a}$ is an ideal of $\mathfrak{h}$. Consider the composition of an inclusion map followed by the quotient map

$$\pi : \mathfrak{b} \hookrightarrow \mathfrak{h} \to \mathfrak{h}/\mathfrak{a}.$$

This is a Lie algebra homomorphism (since both inclusion and quotient are Lie algebra homomorphisms). π is also surjective (every coset $\overline{X} \in \mathfrak{h}/\mathfrak{a}$ can be represented by $X \in \mathfrak{b}$). By (a), we only need to compute $\ker\pi$. If $X \in \mathfrak{b}$, then $\pi(X) = 0$ if and only if $X \in \mathfrak{a}$. Hence $\ker\pi = \mathfrak{a} \cap \mathfrak{b}$.

□

Definition 1.14. *Suppose we have a Lie algebra over F and K/F is a field extension, then we can equip a Lie algebra structure on the K-vector space*

$$\mathfrak{g}^K := \mathfrak{g} \otimes_F K$$

via the rule

$$[X \otimes_F a, Y \otimes_F b] := [X, Y] \otimes_F ab.$$

In the case that $F = \mathbb{R}, K = \mathbb{C}$, then $\mathfrak{g}^{\mathbb{C}}$ is called the ***complexification*** *of the real Lie algebra $\mathfrak{g}$.*

Definition 1.15. *Suppose we have a Lie algebra $\mathfrak{g}$ over K and K/F is a field extension, then we can regard $\mathfrak{g}$ as a Lie algebra over F by restriction of scalar. In the case when $F = \mathbb{R}, K = \mathbb{C}$, then $\mathfrak{g}$ (when regarded as a real Lie algebra) is called the* ***realification*** *of the complex Lie algebra $\mathfrak{g}$.*

Proposition 1.16 ([Kna02, Proposition 1.17])**.** *Suppose $\mathfrak{g}$ is a Lie algebra over F and K/F is a field extension, then for any ideal $\mathfrak{h}$ of $\mathfrak{g}$, we have*

$$[\mathfrak{g}^K, \mathfrak{h}^K] = [\mathfrak{g}, \mathfrak{h}]^K.$$

Proof. As $\mathfrak{g} \subseteq \mathfrak{g}^K, \mathfrak{h} \subseteq \mathfrak{h}^K$, we have $[\mathfrak{g}, \mathfrak{h}] \subseteq [\mathfrak{g}^K, \mathfrak{h}^K]$ as an F-vector space. Thus, by extension of scalar, $[\mathfrak{g}, \mathfrak{h}]^K \subseteq [\mathfrak{g}^K, \mathfrak{h}^K]$ as a K-vector space. Conversely, $[\mathfrak{g}^K, \mathfrak{h}^K]$ is generated by $[X \otimes a, Y \otimes b]$ with $X \in \mathfrak{g}, Y \in \mathfrak{h}$ and $a, b \in K$. Then the proposition follows directly as

$$[X \otimes a, Y \otimes b] = [X, Y] \otimes ab \in [\mathfrak{g}, \mathfrak{h}]^K.$$

□

Example 1.17. *Let $\mathfrak{g}$ be a real Lie algebra of $n \times n$ complex matrices such that for any nonzero $X \in \mathfrak{g}$, $\mathbf{i}X \notin \mathfrak{g}$. Then by we have an injective Lie algebra homomorphism*

$$\iota : \mathfrak{g}^{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C} \to \mathfrak{gl}_n(\mathbb{C})$$

that maps

$$X \otimes 1 \mapsto X, \quad X \otimes \mathbf{i} \mapsto \mathbf{i}X, \quad \forall X \in \mathfrak{g}.$$

Indeed, let $X_1, X_2, \cdots, X_m$ be a basis of $\mathfrak{g}$ (as a real vector space). Then $X_1 \otimes 1, X_2 \otimes 1, \cdots, X_m \otimes 1, X_1 \otimes \mathbf{i}, X_2 \otimes \mathbf{i}, \cdots, X_m \otimes \mathbf{i}$ form a basis of $\mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$. If we have an element

$$X = \sum_{j=1}^{m} a_j X_j \otimes 1 + b_j X_j \otimes \mathbf{i}, \qquad a_j, b_j \in \mathbb{R}$$

such that $\iota(X) = 0$ Then clearly we have

$$\sum_{j=1}^{m} (a_j + \mathbf{i} b_j) X_j = 0.$$

Thus,

$$\sum_{j=1}^{m} a_j X_j = -\mathbf{i} \cdot \sum_{j=1}^{m} b_j X_j \in \mathfrak{g} \cap \mathbf{i}\mathfrak{g}.$$

Hence both the left-hand-side and the right-hand-side are zero. Then the $\mathbb{R}$-linear independence of all X_j forces $a_j = b_j = 0$.

Example 1.18. *For instance, the complexification of* $\mathfrak{sl}_2(\mathbb{R})$ *and* $\mathfrak{su}(2)$ *are both* $\mathfrak{sl}_2(\mathbb{C})$*. Indeed, obviously (by splitting the real and imaginary parts)*

$$\mathfrak{sl}_2(\mathbb{C}) = \mathfrak{sl}_2(\mathbb{R}) \oplus \mathbf{i} \cdot \mathfrak{sl}_2(\mathbb{R}).$$

(This is only a decomposition of real vector space, not a real Lie algebra, simply because $\mathbf{i}\mathfrak{sl}_2(\mathbb{R})$ *is not a real Lie algebra.) Less obviously, For every* $X \in \mathfrak{sl}_2(\mathbb{C})$*, we have*

$$X = \frac{X - \overline{X}^t}{2} + \frac{X + \overline{X}^t}{2}.$$

Then $\frac{X-\overline{X}^t}{2} \in \mathfrak{su}(2)$ *and* $\frac{X+\overline{X}^t}{2} \in \mathbf{i} \cdot \mathfrak{su}(2)$*. In short, we still have a vector space decomposition*

$$\mathfrak{sl}_2(\mathbb{C}) = \mathfrak{su}(2) \oplus \mathbf{i} \cdot \mathfrak{su}(2).$$

We end this example with a further remark. The term 'complex conjugation' is sometimes ambiguous. The complex conjugation relies on the splitting of a complex vector space (regarded as a real vector space by restriction of scalar) into two copies of real vector spaces. Thus, when speaking of complex conjugation, we need to specify with respect to which vector space, we are performing the complex conjugation. For example, given $X = \begin{pmatrix} 1 & \mathbf{i} \\ \mathbf{i} & -1 \end{pmatrix}$*, the complex conjugation with respect to* $\mathfrak{sl}_2(\mathbb{R})$ *is obtained by the usual conjugation*

$$\overline{X} = \begin{pmatrix} 1 & -\mathbf{i} \\ -\mathbf{i} & -1 \end{pmatrix}.$$

Yet if we conjugate with respect to $\mathfrak{su}(2)$*, the conjugate of* X *is*

$$-\overline{X}^t = -\begin{pmatrix} 1 & -\mathbf{i} \\ -\mathbf{i} & -1 \end{pmatrix}.$$

Example 1.19. $\mathfrak{so}(2,1)$ *is isomorphic to* $\mathfrak{sl}_2(\mathbb{R})$ *as real Lie algebras (Homework).*

Example 1.20. *In this example, we show that* $\mathfrak{su}(2)$ *is isomorphic to* $\mathfrak{so}(3)$ *as real Lie algebras.*

Take a basis of $\mathfrak{su}(2)$

$$E_1 = \frac{1}{2}\begin{pmatrix} \mathbf{i} & \\ & -\mathbf{i} \end{pmatrix}, \quad E_2 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad E_3 = \frac{1}{2}\begin{pmatrix} 0 & \mathbf{i} \\ \mathbf{i} & 0 \end{pmatrix}.$$

Then one can check that

$$[E_1, E_2] = E_3, \quad [E_2, E_3] = E_1, \quad [E_3, E_1] = E_2.$$

Let us also take a basis of $\mathfrak{so}(3)$

$$X_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, X_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Then one can check that

$$[X_1, X_2] = X_3, \quad [X_2, X_3] = X_1, \quad [X_3, X_1] = X_2.$$

This shows that $E_j \mapsto X_j$ *provides a Lie algebra isomorphism* $\mathfrak{su}(2) \simeq \mathfrak{so}(3)$*.*

Remark 1.21. *A better (conceptual) proof will be discussed later after we introduce the adjoint representation of a Lie group.*

2. Solvable and Nilpotent Lie Algebras

We assume that $\mathfrak{g}$ is a finite-dimensional Lie algebra in this Subsection.

Definition 2.1. *Let $\mathfrak{g}$ be a Lie algebra over F. We defined inductively that $\mathfrak{g}^0 := \mathfrak{g}, \mathfrak{g}^1 := [\mathfrak{g}, \mathfrak{g}], \mathfrak{g}^{j+1} := [\mathfrak{g}^j, \mathfrak{g}^j]$. Then we call*

$$\mathfrak{g} = \mathfrak{g}^0 \supseteq \mathfrak{g}^1 \supseteq \mathfrak{g}^2 \cdots$$

*the **commutator series** of $\mathfrak{g}$. We say $\mathfrak{g}$ is **solvable** if the commutator series terminates.*

Definition 2.2. *Let $\mathfrak{g}$ be a Lie algebra over F. We defined inductively that $\mathfrak{g}_0 := \mathfrak{g}, \mathfrak{g}_1 := [\mathfrak{g}, \mathfrak{g}], \mathfrak{g}_{j+1} := [\mathfrak{g}, \mathfrak{g}_j]$. Then we call*

$$\mathfrak{g} = \mathfrak{g}_0 \supseteq \mathfrak{g}_1 \supseteq \mathfrak{g}_2 \cdots$$

*the **lower central series** of $\mathfrak{g}$. We say $\mathfrak{g}$ is **nilpotent** if the lower central series terminates.*

Example 2.3. *Consider the standard maximal nilpotent subalgebra $\mathfrak{n}$ of $\mathfrak{gl}_4(F)$. Explicitly, $\mathfrak{n}$ consists of all strictly upper triangular matrices in $\mathfrak{gl}_4(F)$. (Hmm..... I realize I am abusing the terminology here. We of course need to check that $\mathfrak{n}$ is nilpotent, but this is what we exactly are about to do.) Then*

$$\mathfrak{n}_1 = [\mathfrak{n}, \mathfrak{n}] = \left\{ \begin{pmatrix} 0 & 0 & \star & \star \\ 0 & 0 & 0 & \star \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \right\},$$

$$\mathfrak{n}_2 = [\mathfrak{n}_1, \mathfrak{n}] = \left\{ \begin{pmatrix} 0 & 0 & 0 & \star \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \right\},$$

$$\mathfrak{n}_3 = [\mathfrak{n}_2, \mathfrak{n}] = 0.$$

Example 2.4. *Consider the standard Borel subalgebra $\mathfrak{b}$ of $\mathfrak{gl}_4(F)$. Explicitly, $\mathfrak{b}$ consists of all upper triangular matrices in $\mathfrak{gl}_4(F)$. (Surely, it is better to give a coordinate-free definition of Borel subalgebra. Yet I guess it is better to exhibit examples rather than spending too much time on abstract theory.) Then*

$$\mathfrak{b}_1 = [\mathfrak{b}, \mathfrak{b}] = \mathfrak{n}$$

defined in the Example 2.3 (the standard maximal unipotent subalgebra of $\mathfrak{gl}_4(F)$).

$$\mathfrak{b}_2 = [\mathfrak{b}_1, \mathfrak{b}_1] = [\mathfrak{n}, \mathfrak{n}] = \mathfrak{n}_1$$

defined in the Example 2.3.

$$\mathfrak{b}_3 = [\mathfrak{b}_2, \mathfrak{b}_2] = [\mathfrak{n}_1, \mathfrak{n}_1] = 0.$$

(Be careful here! $[\mathfrak{n}_1, \mathfrak{n}] \neq 0$, yet $[\mathfrak{n}_1, \mathfrak{n}_1] = 0$.)

Example 2.5. *Let $a, b, c, d \in F$ with $ad - bc \neq 0$. Consider the three dimensional Lie algebra $\mathfrak{g}$ with basis X, Y, Z satisfying*

$$[X, Y] = 0, [X, Z] = aX + bY, [Y, Z] = cX + dY.$$

Then X, Y spans a two dimensional abelian Lie subalgebra of $\mathfrak{g}$, denoted by $\mathfrak{h}$. Moreover, $\mathfrak{h}$ is an ideal of $\mathfrak{g}$. We have the commutator series

$$\mathfrak{g} \supseteq [\mathfrak{g}, \mathfrak{g}] = \mathfrak{h} \supseteq 0 = [\mathfrak{h}, \mathfrak{h}].$$

This shows that $\mathfrak{g}$ is solvable. As for the lower central series, since $\mathfrak{h}$ is a non zero ideal of $\mathfrak{g}$, the lower central series will not terminate. Hence $\mathfrak{g}$ is not nilpotent.

Now we enforce $b = c = 0, d = 1, F = \mathbb{R}$. We aim to show that when a varies in $\mathbb{R}$, all these three dimensional Lie algebra $\mathfrak{g}$ are non-isomorphic. Indeed, we consider the adjoint representation of $\mathfrak{g}$ on the two dimensional ideal $\mathfrak{h} = [\mathfrak{g}, \mathfrak{g}]$. As $\mathfrak{h}$ is abelian, this descends to a representation of $\mathfrak{g}/\mathfrak{h}$ on $\mathfrak{h}$, denoted by τ. As $\mathfrak{g}/\mathfrak{h}$ is a one-dimensional abelian Lie algebra, this representation is characterized by two weights, or equivalently, a vector $(a, 1)$ in the projective line. Thus, when a varies, these three dimensional Lie algebras are non-isomorphic.

Example 2.6. *The $(2n+1)$-dimensional Heisenberg Lie algebra (over $\mathbb{R}$), denoted by $\mathfrak{h}_n$, is a Lie subalgebra of $\mathfrak{sp}_{2n+2}(\mathbb{R})$:*

$$\mathfrak{h}_n = \left\{ \begin{pmatrix} 0 & x & y & z \\ 0 & 0 & 0 & y^t \\ 0 & 0 & 0 & -x^t \\ 0 & 0 & 0 & 0 \end{pmatrix} \middle| \, x, y \in \mathbb{R}^n \text{ are row vectors} \right\}.$$

It is easy to see that $[\mathfrak{h}_n, \mathfrak{h}_n] \simeq \mathbb{R}$ is the center of $\mathfrak{h}_n$, denoted by $\mathfrak{z}_n$ at the moment. Hence $\mathfrak{h}_n$ is a two-step nilpotent Lie algebra. Then, we have an exact sequence

$$0 \to \mathbb{R} \to \mathfrak{h}_n \to \mathbb{R}^{2n} \to 0,$$

meaning that $\mathfrak{h}_{2n+1}$ is a nontrivial central extension of $\mathbb{R}^{2n}$.

Suppose that $\mathfrak{b}$ is any non-abelian Lie subalgebra of $\mathfrak{h}_n$, then

$$[\mathfrak{b}, \mathfrak{b}] \subseteq [\mathfrak{h}_n, \mathfrak{h}_n] = \mathfrak{z}_n.$$

Thus, any non-abelian Lie subalgebra must contain $\mathfrak{z}_n$. Moreover, if $\mathfrak{b}$ contains the center $\mathfrak{z}_n$, then

$$[\mathfrak{b}, \mathfrak{h}_n] \subseteq [\mathfrak{h}_n, \mathfrak{h}_n] = \mathfrak{z}_n \subseteq \mathfrak{b},$$

and hence $\mathfrak{b}$ must be an ideal of $\mathfrak{h}_n$.

In particular, when $n = 1$, the only Lie subalgebras of $\mathfrak{h}_1$ are

1. *$\{0\}$ (the zero-dimensional trivial Lie algebra);*
2. *one dimensional abelian Lie subalgebras;*
3. *two dimensional Lie subalgebras (which contains the center automatically, hence are abelian);*
4. *the whole Lie algebra $\mathfrak{h}_1$.*

The ideals of $\mathfrak{h}_1$ are $\{0\}$, the center $\mathfrak{z}_1$, all two dimensional Lie subalgebras and $\mathfrak{h}_1$ itself.

Remark 2.7. *Central extension of a Lie algebra is related to the second Lie algebra cohomology, which will be discussed in future lectures. The Heisenberg Lie algebra $\mathfrak{h}_1$ is also isomorphic to the standard maximal nilpotent subalgebra of $\mathfrak{gl}_3(\mathbb{R})$. A variance of this isomorphism is discussed in Example 5.7.*

Example 2.8. *Consider $V := \mathbb{C}[x]$. Let*

$$\widetilde{X} \in \mathrm{End}(V), \quad f(x) \mapsto \frac{\partial}{\partial x} f(x)$$

and

$$\widetilde{Y} \in \mathrm{End}(V), \quad f(x) \mapsto x f(x).$$

The associative algebra $\mathrm{End}(V)$ *naturally has a Lie algebra structure. Moreover, it is easy to check that*

$$[\widetilde{X}, \widetilde{Y}] = \mathrm{Id}_V$$

and the identity operator Id_V *lies in the center of* $\mathrm{End}(V)$*. Thus, we have a realization of the Heisenberg Lie algebra* $\mathfrak{h}_1^{\mathbb{C}}$ *as a subalgebra of* $\mathrm{End}(V)$*. The Lie algebra embedding is given by*

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \mapsto \widetilde{X}, \quad \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \mapsto \widetilde{Y}, \quad \begin{pmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \mapsto \mathrm{Id}_V.$$

In general, we may consider $V = \mathbb{C}[x_1, x_2, \cdots, x_n]$*. Set*

$$\widetilde{X}_k := \frac{\partial}{\partial x_k} \quad \textit{differentiation}$$

$$\widetilde{Y}_k := x_k \quad \textit{multiplication}$$

Then we have a realization of $\mathfrak{h}_n^{\mathbb{C}}$ *as a subalgebra of* $\mathrm{End}(V)$*.*

We now aim to justify that all solvable Lie algebras should look like subalgebras of $\mathfrak{b}$ in Example 2.4 and all nilpotent Lie algebras should look like subalgebras of $\mathfrak{n}$ in Example 2.3. We start from a few simple facts.

Lemma 2.9. *All* $\mathfrak{g}^j$ *and* $\mathfrak{g}_j$ *are automatically ideals of* $\mathfrak{g}$*.*

Proof. Obviously, $\mathfrak{g}^1 = \mathfrak{g}_1$ is an ideal of $\mathfrak{g}$. By induction,

$$[\mathfrak{g}^{j+1}, \mathfrak{g}] = [[\mathfrak{g}^j, \mathfrak{g}^j], \mathfrak{g}] \subseteq [[\mathfrak{g}^j, \mathfrak{g}], \mathfrak{g}^j] + [[\mathfrak{g}, \mathfrak{g}^j], \mathfrak{g}^j] \subseteq [\mathfrak{g}^j, \mathfrak{g}^j] + [\mathfrak{g}^j, \mathfrak{g}^j] = \mathfrak{g}^{j+1};$$

and

$$[\mathfrak{g}_{j+1}, \mathfrak{g}] = [[\mathfrak{g}_j, \mathfrak{g}], \mathfrak{g}] \subseteq [[\mathfrak{g}, \mathfrak{g}], \mathfrak{g}_j] + [[\mathfrak{g}, \mathfrak{g}_j], \mathfrak{g}] \subseteq [\mathfrak{g}, \mathfrak{g}_j] + [\mathfrak{g}, \mathfrak{g}_j] = \mathfrak{g}_{j+1}.$$

□

Lemma 2.10. *Clearly all nilpotent Lie algebras are solvable.*

Proof. This is because $\mathfrak{g}_{j+1} = [\mathfrak{g}_j, \mathfrak{g}] \supset \mathfrak{g}^{j+1}$ by induction. Thus $\mathfrak{g}_j = 0$ forces $\mathfrak{g}^j = 0$. □

Lemma 2.11. *For any* $\mathfrak{g}$ *(not necessarily solvable),* $\mathfrak{g}^j/\mathfrak{g}^{j+1}$ *is abelian.*

Proof. Set $\mathfrak{h} = \mathfrak{g}^j$ and $\mathfrak{b} = [\mathfrak{h}, \mathfrak{h}]$ Then we have a natural surjective Lie algebra homomorphism:

$$\pi : \mathfrak{h} \to \mathfrak{h}/\mathfrak{b}.$$

Clearly, π sends $[\mathfrak{h}, \mathfrak{h}]$ to 0. Thus,

$$[\mathfrak{h}/\mathfrak{b}, \mathfrak{h}/\mathfrak{b}] = \pi([\mathfrak{h}, \mathfrak{h}]) = 0.$$

The lemma follows. □

Proposition 2.12 ([Kna02, Proposition 1.10, 1.11, 1.17])**.** (1) *Any subalgebra or homomorphic image of a solvable (nilpotent resp.) Lie algebra is solvable (nilpotent resp.).*

(2) *Suppose we have an exact sequence of Lie algebras:*

$$0 \to \mathfrak{a} \to \mathfrak{g} \to \mathfrak{g}/\mathfrak{a} \to 0,$$

then $\mathfrak{g}$ *is solvable if and only if both* $\mathfrak{a}$ *and* $\mathfrak{g}$ *are solvable. In particular,* $\mathfrak{g}$ *is solvable if and only if* $[\mathfrak{g}, \mathfrak{g}]$ *is solvable.*

(3) *Suppose we have an exact sequence of Lie algebras:*

$$0 \to \mathfrak{a} \to \mathfrak{g} \to \mathfrak{g}/\mathfrak{a} \to 0,$$

then $\mathfrak{g}$ *is nilpotent implies that both* $\mathfrak{a}$ *and* $\mathfrak{g}$ *are nilpotent.*

(4) *Suppose* $\mathfrak{g}$ *is a finite-dimensional Lie algebra over* F *and* K/F *is a finite extension. Then* $\mathfrak{g}$ *is solvable (resp. nilpotent) if and only if* $\mathfrak{g}^K$ *is solvable (resp. nilpotent).*

Proof. 1) If $\mathfrak{h}$ is a subalgebra of $\mathfrak{g}$, then a simple induction gives $\mathfrak{h}^j \subseteq \mathfrak{g}^j$ (resp. $\mathfrak{h}_j \subseteq \mathfrak{g}_j$). Hence $\mathfrak{g}$ solvable (resp. nilpotent) implies $\mathfrak{h}$ solvable (resp. nilpotent). If $\pi : \mathfrak{g} \to \mathfrak{h}$ is a surjective homomorphism, then $\pi(\mathfrak{g}^j) = \mathfrak{h}^j$ (resp. $\pi(\mathfrak{g}_j) = \mathfrak{h}_j$). Hence $\mathfrak{g}$ solvable (resp. nilpotent) implies $\mathfrak{h}$ solvable (resp. nilpotent).

2) $\Rightarrow$: follows from 1).

$\Leftarrow$: Suppose that $(\mathfrak{g}/\mathfrak{a})^k = 0$ and $\mathfrak{a}^j = 0$, then $\mathfrak{g}^{k+j} = ((\mathfrak{g})^k)^j \subseteq (\mathfrak{a})^j = 0$. Hence $\mathfrak{g}$ is solvable.

The last statement follows from Lemma 2.11.

3) follows from 1).

4)This follows from Lemma 2.9 and Proposition 1.16. □

Lemma 2.13. *Let* $\mathfrak{g}$ *be a finite-dimensional Lie algebra over* F *and* $\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$. *Then we have an exact sequence*

$$0 \to \mathfrak{z}_{\mathfrak{g}} \to \mathfrak{g} \to \mathrm{ad}(\mathfrak{g}) \to 0.$$

As a consequence, $\mathfrak{g}$ *is solvable (resp. nilpotent) if and only if* $\mathrm{ad}(\mathfrak{g})$ *is solvable (resp. nilpotent).*

Proof. The exact sequence is obvious. The last statement follows from Proposition 2.12. □

Proposition 2.14 ([Kna02, Proposition 1.23])**.** *An* n*-dimensional Lie algebra* $\mathfrak{g}$ *is solvable if and only if it has an* ***elementary sequence****, i.e. a decreasing sequence of subalgebras*

$$\mathfrak{g} = \mathfrak{a}_0 \supseteq \mathfrak{a}_1 \supseteq \mathfrak{a}_2 \cdots \supseteq \mathfrak{a}_n = 0$$

such that $\mathfrak{a}_{j+1}$ *is an ideal of* $\mathfrak{a}_j$ *of codimension 1.*

Proof. Let $\mathfrak{g}$ be solvable. Interpolate subspaces $\mathfrak{a}_i$ in the commutator series so that

$$\dim(\mathfrak{a}_i/\mathfrak{a}_{i+1}) = 1$$

for all i. For any i, we can find a unique j such that $\mathfrak{g}^{j+1} \subseteq \mathfrak{a}_{i+1} \subseteq \mathfrak{a}_i \subseteq \mathfrak{g}^j$. Then

$$[\mathfrak{a}_i, \mathfrak{a}_{i+1}] \subseteq [\mathfrak{g}^j, \mathfrak{g}^j] \subseteq \mathfrak{g}^{j+1} \subseteq \mathfrak{a}_{i+1}.$$

Hence $\mathfrak{a}_i$ is subalgebra for each i, and $\mathfrak{a}_{i+1}$ is an ideal in $\mathfrak{a}_i$.

Conversely, assume that the sequence exists. We show by induction that $\mathfrak{g}^j \subseteq \mathfrak{a}_j$. Choose x_i, so that $\mathfrak{a}_i = F \cdot x_i + \mathfrak{a}_{i+1}$. Then

$$\mathfrak{g}^{j+1} = [\mathfrak{g}^j, \mathfrak{g}^j] \subseteq [\mathfrak{a}_j, \mathfrak{a}_j] \subseteq [F \cdot x_j + \mathfrak{a}_{j+1}, F \cdot x_j + \mathfrak{a}_{j+1}] \subseteq [F \cdot x_j, \mathfrak{a}_{j+1}] + [\mathfrak{a}_{j+1}, \mathfrak{a}_{j+1}] \subseteq \mathfrak{a}_{j+1}.$$

Thus, $\mathfrak{g}^n \subseteq \mathfrak{a}_n = 0$. □

Remark 2.15. *The spirit of this proof imitates the proof of the structure of finite solvable groups. For finite solvable groups, interpolation leads to a subnormal series with factor groups of prime order ('arguably' the simplest group). Here interpolation leads to a series with factor algebra being 'arguably' the simplest (one-dimensional abelian Lie algebra).*

3. Engel's Theorem

Recall the definition of a Lie algebra representation:

Definition 3.1. *We say (π, V), where V is an F-vector space, is a* ***representation*** *of a Lie algebra $\mathfrak{g}$ (defined over F) if π is a Lie algebra homomorphism from $\mathfrak{g}$ to $\mathrm{End}_F(V)$. From this, we can define* ***subrepresentation*** *and* ***quotient representation****.*

Remark 3.2. *For example, when we say (π, V_π) is a $\mathfrak{gl}_n(\mathbb{R})$-module, we regard the vector space V_π as a real vector space (though probably it is complex in prior) and $\pi : \mathfrak{gl}_n(\mathbb{R}) \to \mathrm{End}_{\mathbb{R}}(V)$ is only $\mathbb{R}$-linear.*

Before discussing Engel's theorem, we look at the following *trivial* example.

Example 3.3. *Take $\mathfrak{g} = \mathbb{C}$. It is abelian (hence nilpotent and solvable). Consider any nonzero linear function*

$$\pi : \mathfrak{g} \to \mathbb{C},$$

as $\mathfrak{g} = \mathbb{C}$ is abelian,

$$\pi([X, Y]) = 0 = [\pi(X), \pi(Y)].$$

Thus, π can be viewed as a representation of $\mathfrak{g}$ on $\mathbb{C}$ ($\mathrm{End}_{\mathbb{C}}(\mathbb{C}) = \mathbb{C}$). As a 1×1 matrix, $\pi(\mathfrak{g})$ is upper triangular, but not necessarily strictly upper triangular (i.e. not necessarily identically zero).

Definition 3.4. *A finite dimensional representation π of $\mathfrak{g}$ is said to be* ***nilpotent*** *if for all $X \in \mathfrak{g}$, $\pi(X)$ is a nilpotent operator (i.e. a nilpotent matrix under any basis).*

Theorem 3.5 (Engel's Theorem,[Kna02, Theorem 1.35])**.** *Let $\mathfrak{g}$ be a Lie algebra over F and (π, V) is a finite dimensional nilpotent representation of $\mathfrak{g}$.*

1. *Then $V^{\mathfrak{g}} := \{v \in V \mid \pi(X)v = 0, \forall X \in \mathfrak{g}\}$ is nonzero.*
2. *There exists a basis of V such that $\pi(\mathfrak{g})$ is the subalgebra of the standard maximal nilpotent subalgebra of $\mathrm{End}_F(V)$ (which consists all strictly upper triangular matrices).*

Proof of Theorem 3.5 (1) when $\mathfrak{g}$ is nilpotent. We prove 1) by induction on the dimension of $\mathfrak{g}$. Since $\mathfrak{g}$ is nilpotent, hence $[\mathfrak{g}, \mathfrak{g}] \neq \mathfrak{g}$. We take a codimensional one subspace $\mathfrak{h}$ of $\mathfrak{g}$ containing $[\mathfrak{g}, \mathfrak{g}]$. Then

$$[\mathfrak{h}, \mathfrak{g}] \subseteq [\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{h},$$

i.e. $\mathfrak{h}$ is an ideal of $\mathfrak{g}$, hence nilpotent (Proposition 2.12). Thus, by induction hypothesis $V^{\mathfrak{h}} \neq 0$.

Now we check $\pi(X_0)$ stabilizes $V^{\mathfrak{h}}$. Take $X_0 \in \mathfrak{g}$ such that $\mathfrak{g} = F \cdot X_0 + \mathfrak{h}$. Then for any $v \in V^{\mathfrak{h}}$ and $H \in \mathfrak{h}$,

$$\pi(H)\pi(X_0)v = \pi(X_0)\pi(H)v + \pi([H, X_0])v = 0.$$

Thus, $V^{\mathfrak{h}}$ is $\pi(X_0)$-stable. As $\pi(X_0)$ acts nilpotently on $V^{\mathfrak{h}}$, there exists a nonzero vector $v \in V^{\mathfrak{h}}$ annihilated by $\pi(X_0)$. Therefore $v \in V^{\mathfrak{g}}$. □

Proof of Theorem 3.5 (1), general case. From the proof of Theorem 3.5 in the nilpotent case, we see that the key point for the induction is to construct a codimensional one nilpotent ideal $\mathfrak{h}$ of $\mathfrak{g}$. (This step is quite straightforward when $\mathfrak{g}$ is nilpotent.) Yet this condition may fail when $\mathfrak{g}$ is not nilpotent. Instead, we aim to find a codimensional one nilpotent ideal of $\mathfrak{h}$ in $\pi(\mathfrak{g})$. In such a general case, we need a bit more work.

The proof is based on an induction on the dimension of $\pi(\mathfrak{g})$ (instead of $\mathfrak{g}$), the dimension of the Lie subalgebra of $\mathrm{End}_F(V)$. Take a maximal proper subalgebra $\mathfrak{h}$ of $\pi(\mathfrak{g})$. This subalgebra $\mathfrak{h}$ is nilpotent by inductive hypothesis. Then by a direct computation

$$(\mathrm{ad}(H))^{2m}(\pi(X)) = \sum_{j=0}^{2m}(-1)^j \begin{pmatrix} 2m \\ j \end{pmatrix} H^j \pi(X) H^{2m-j}.$$

As $H \in \pi(\mathfrak{g})$ is nilpotent, we see that $\mathrm{ad}(H)$ acts nilpotently on $\pi(\mathfrak{g})$. Hence $\mathrm{ad}(H)$ acts nilpotently on the quotient space $\pi(\mathfrak{g})/\mathfrak{h}$.

By inductive hypothesis (since $\mathfrak{h}$ has smaller dimension), the adjoint representation of $\mathfrak{h}$ on $\pi(\mathfrak{g})/\mathfrak{h}$ has a simultaneous eigenvector. In other words, we can find a representative $X_0 \in \pi(\mathfrak{g})$ such that

$$[\mathfrak{h}, \overline{X}_0] = 0.$$

This implies that $\mathfrak{h} + F \cdot X_0$ is a subalgebra of $\pi(\mathfrak{g})$ strictly containing $\mathfrak{h}$. By maximality of $\mathfrak{h}$, we see that $\mathfrak{h} + F \cdot X_0 = \pi(\mathfrak{g})$. Then $\mathfrak{h}$ is a codimensional nilpotent ideal of $\pi(\mathfrak{g})$. The rest of the proof is the same as that of the nilpotent case. □

Proof of Theorem 3.5 (2). The proof is based on an induction on $\dim V$. The case when $\dim V = 1$ is trivial.

Now we suppose that the corollary holds for all V with dimension less than n. When $\dim V = n$, by Engel's Theorem, any nonzero $e_n \in V^{\mathfrak{g}}$ is a simultaneous eigenvector with zero eigenvalue. The space $W := V/(F \cdot e_n)$ is a representation of $\mathfrak{g}$ with dimension less then n. Then we choose a basis $\widetilde{e}_1, \widetilde{e}_2, \cdots, \widetilde{e}_{n-1}$ such that under such a basis, $\mathfrak{g}$-action on W is strictly upper triangular. Take any preimage e_j of $\widetilde{e}_j$. By construction, under $e_1, e_2, \cdots, e_n$, $\pi(\mathfrak{g})$ is strictly upper triangular. □

Corollary 3.6 (Engel's Theorem for Adjoint Representation)**.** *The Lie algebra $\mathfrak{g}$ is nilpotent if and only if $\mathrm{ad}(X)$ is a nilpotent operator for each $X \in \mathfrak{g}$.*

Proof. $\mathfrak{g}$ is nilpotent if and only if $\mathrm{ad}\mathfrak{g}$ is nilpotent (Lemma 2.13), if and only if $\mathrm{ad}(X)$ is nilpotent for all $X \in \mathfrak{g}$ (Theorem 3.5). □

Remark 3.7. *When $\mathfrak{g}$ is nilpotent, by taking $V = \mathfrak{g}$, one can find a suitable basis of $\mathfrak{g}$ such that the adjoint representation of $\mathfrak{g}$ on itself maps $\mathfrak{g}$ into the standard maximal nilpotent subalgebra of $\mathrm{End}_F(\mathfrak{g})$.*

4. Lie's Theorem

Theorem 4.1 (Lie Theorem, [Kna02, Theorem 1.25])**.** *Let (π, V) be a finite dimensional nonzero representation of a finite-dimensional solvable Lie algebra $\mathfrak{g}$, where V and $\mathfrak{g}$ are vector spaces over F). If all the eigenvalues of $\pi(\mathfrak{g})$ lie in F, then there exists a simultaneous eigenvector for all $\pi(X)$ with $X \in \mathfrak{g}$.*

Proof. An elementary (but long) proof can be found in Knapp's book and I will not bother to copy it down here. The proof is a bit more complicated than that of Engel's theorem, because we lose control on the eigenvalues of $\pi(\mathfrak{g})$ (in nilpotent case, we only have one eigenvalue). Thus, a filtration on the representation space is needed. This filtration in fact comes from the filtration of the universal enveloping algebra. Students are encouraged to revisit the proof after learning universal enveloping algebra. □

Corollary 4.2. *Retain the notation and assumptions of Theorem 4.1. Then, one can choose the basis of V such that π is in fact a Lie algebra homomorphism from $\mathfrak{g}$ to the standard upper* ***Borel subalgebra*** *(the subalgebra containing all upper triangular matrices) of* $\mathrm{End}_F(V)$.

Proof. The proof is based on an induction on $\dim V$. The case when $\dim V = 1$ is trivial.

Now we suppose that the corollary holds for all V with dimension less than n. When $\dim V = n$, by Theorem 4.1, there exists a simultaneous eigenvector e_n of $\pi(\mathfrak{g})$. The space $W := V/(F \cdot e_n)$ is a representation of $\mathfrak{g}$ with dimension less then n. Then we choose a basis $\widetilde{e}_1, \widetilde{e}_2, \cdots, \widetilde{e}_{n-1}$ such that under such a basis, $\mathfrak{g}$-action on W is upper triangular. Take any preimage e_j of $\widetilde{e}_j$. By construction, under $e_1, e_2, \cdots, e_n$, $\pi(\mathfrak{g})$ is upper triangular. □

Remark 4.3. *To describe the action of $\mathfrak{g}$ on V, we usually use* ***invariant flags****, i.e. a decreasing sequence of $\mathfrak{g}$-invariant spaces*

$$V = V_0 \supseteq V_1 \supseteq V_2 \supseteq \cdots \supseteq V_n = 0$$

such that V_{j+1} is a codimensional one subspace of V_j. Then V_j is exactly the F-span of $e_n, e_{n-1}, \cdots, e_{n-j+1}$.

5. Radical and Nilpotent Radical

Proposition 5.1 ([Kna02, Proposition 1.12, Proposition 1.14])**.** *The sum of two solvable Lie ideals of $\mathfrak{g}$ is still a solvable ideal. Thus there exists a unique maximal solvable ideal of $\mathfrak{g}$, which is called the* ***radical*** *of $\mathfrak{g}$ and denoted by* $\mathrm{rad}(\mathfrak{g})$.

Proof. Suppose that $\mathfrak{a}, \mathfrak{b}$ are solvable ideals, then $(\mathfrak{a}+\mathfrak{b})/\mathfrak{b} \simeq \mathfrak{a}/(\mathfrak{a}\cap\mathfrak{b})$ is solvable (Proposition 2.12), and hence $\mathfrak{a} + \mathfrak{b}$ is solvable (by Proposition 2.12 again). □

Proposition 5.2 ([Var84, Theorem 3.8.1])**.** *Suppose that K/F is a field extension, then* $\mathrm{rad}(\mathfrak{g}^K) = (\mathrm{rad}(\mathfrak{g}))^K$. *If $\mathfrak{h}$ is an ideal, so is* $\mathrm{rad}(\mathfrak{h})$. *Moreover,* $\mathrm{rad}(\mathfrak{h}) = \mathrm{rad}(\mathfrak{g}) \cap \mathfrak{h}$.

Proof. Omitted. □

Proposition 5.3 ([Kna02, Proposition 1.39, 1.40])**.**
1. *For a solvable Lie algebra $\mathfrak{g}$, $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.*
2. *For a solvable Lie algebra $\mathfrak{g}$, it has a unique maximal nilpotent ideal $\mathfrak{n}$, called the* ***nilpotent radical*** *or* ***nilradical*** *of $\mathfrak{g}$.*

Proof. To prove 1), we take $K = \overline{F}$, an algebraic closure of F. We apply Lie's theorem with $V = \mathfrak{g}^K, \pi = \mathrm{ad}$. Then $\mathrm{ad}\mathfrak{g}^K$ are upper triangular under a certain basis of $\mathfrak{g}^K$. This shows that $\mathrm{ad}[\mathfrak{g}^K, \mathfrak{g}^K]$ is nilpotent. Hence $[\mathfrak{g}^K, \mathfrak{g}^K]$ is nilpotent (Lemma 2.13). Then $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent (Proposition 2.12).

To prove 2), we take $K = \overline{F}$, an algebraic closure of F. We apply Lie's theorem with $V = \mathfrak{g}^K, \pi = \mathrm{ad}$ again. Regard $\mathfrak{g}$ as a Lie subalgebra of $\mathfrak{g}^K$. Then set

$$\mathfrak{n} := \{X \in \mathfrak{g} \,|\, \mathrm{ad}X \text{ is nilpotent}\}.$$

Obviously, this is an ideal of $\mathfrak{g}$ (since $\mathrm{ad}\mathfrak{g}$ are all upper triangular while $\mathrm{ad}\mathfrak{n}$ are all strictly upper triangular). We only need to check that $\mathfrak{n}$ is maximal. Indeed, if $\mathfrak{n}'$ is another nilpotent ideal and $X \in \mathfrak{n}'$, take an arbitrary $Y \in \mathfrak{g}$, then

$$\mathrm{ad}(X)^j(Y) = \mathrm{ad}(X)^{j-1}[X, Y] \in \mathrm{ad}(X)^{j-1}\mathfrak{n}' = 0$$

for sufficiently large j. This shows that $\mathrm{ad}X$ is nilpotent, hence $X \in \mathfrak{n}$. □

Remark 5.4. *In general, for any Lie algebra* $\mathfrak{g}$*, the nilradical of* $\mathrm{rad}(\mathfrak{g})$ *(denoted by* $\mathrm{nilrad}(\mathfrak{g})$*) is also the maximal nilpotent ideal of* $\mathfrak{g}$*. Hence we can define it to be the nilradical of* $\mathfrak{g}$*. Moreover,* $[\mathfrak{g}, \mathrm{rad}(\mathfrak{g})] \subseteq \mathrm{nilrad}(\mathfrak{g})$*. But it needs a bit more work to prove this claim (see* [Kna02, Corollary 1.41]*).*

Proposition 5.5 ([Var84, Theorem 3.8.3])**.** *Suppose that* K/F *is a finite extension, then* $\mathrm{nilrad}(\mathfrak{g}^K) = (\mathrm{nilrad}\mathfrak{g})^K$*. If* $\mathfrak{h}$ *is an ideal of* $\mathfrak{g}$*, so is* $\mathrm{nilrad}(\mathfrak{h})$*. Moreover,* $\mathrm{nilrad}(\mathfrak{h}) = (\mathrm{nilrad}(\mathfrak{g})) \cap \mathfrak{h}$*.*

Lemma 5.6. *Suppose that* $\mathfrak{g}$ *is solvable. Then* $\mathfrak{g}/\mathrm{nilrad}(\mathfrak{g})$ *is abelian (Homework).*

To end this section, we supplement an example below, in view of Proposition 5.3. (In some references, the Heisenberg Lie algebra $\mathfrak{h}_1$ is defined to be the standard maximal nilpotent Lie algebra of $\mathfrak{sl}_3(\mathbb{R})$, which consists of all strictly upper triangular matrices. The following example (implicitly) proves the equivalence of these two definitions.)

Example 5.7. *Let* $\mathfrak{g} := \left\{ \begin{pmatrix} \mathbf{i}\theta & 0 & 0 \\ z & -2\mathbf{i}\theta & 0 \\ \mathbf{i}t & \overline{z} & \mathbf{i}\theta \end{pmatrix} \middle| z \in \mathbb{C}, \theta \in \mathbb{R}, t \in \mathbb{R} \right\}$*. Then* $[\mathfrak{g}, \mathfrak{g}]$ *is isomorphic to the real Heisenberg Lie algebra* $\mathfrak{h}_1$*.*

Indeed, by a direct computation,

$$[\mathfrak{g}, \mathfrak{g}] := \left\{ Z = \begin{pmatrix} 0 & 0 & 0 \\ z & 0 & 0 \\ \mathbf{i}t & \overline{z} & 0 \end{pmatrix} \middle| \right\}.$$

Take a basis of $\mathfrak{h}_1$ *:*

$$X = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Then take a basis of $[\mathfrak{g}, \mathfrak{g}]$*:*

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & 0 \\ \mathbf{i} & 0 & 0 \\ 0 & -\mathbf{i} & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 2\mathbf{i} & 0 & 0 \end{pmatrix}.$$

Then one directly checks that

$$X \mapsto A, \quad Y \mapsto B, \quad Z \mapsto C$$

provides an isomorphism $\mathfrak{h}_1 \simeq [\mathfrak{g}, \mathfrak{g}]$*.*

References

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[Var84] V. S Varadarajan, *Lie groups, Lie algebras, and their representations*, Graduate texts in mathematics 102, New York : Springer-Verlag 1984.

Department of Mathematics, Zhejiang University, China
Email address: tianfangyangmath@zju.edu.cn